

PASSWORD SECURITY AUDIT TOOL

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INTRODUCTION:

The Password Security Audit Tool is a Python-based security tool used to analyze the strength and security of passwords. It checks password length, uppercase and lowercase characters, numbers, special characters, common passwords, estimated crack time, and generates a stronger password suggestion. It also uses bcrypt to securely hash passwords.

OBJECTIVE:

To check the strength of a given password.

To identify common and weak passwords.

To estimate the approximate password cracking time.

To generate a stronger password suggestion.

To securely hash passwords using bcrypt.

To improve awareness of password security.

TOOLS AND TECHNOLOGIES USED:

Programming Language: Python

IDE: Python IDLE

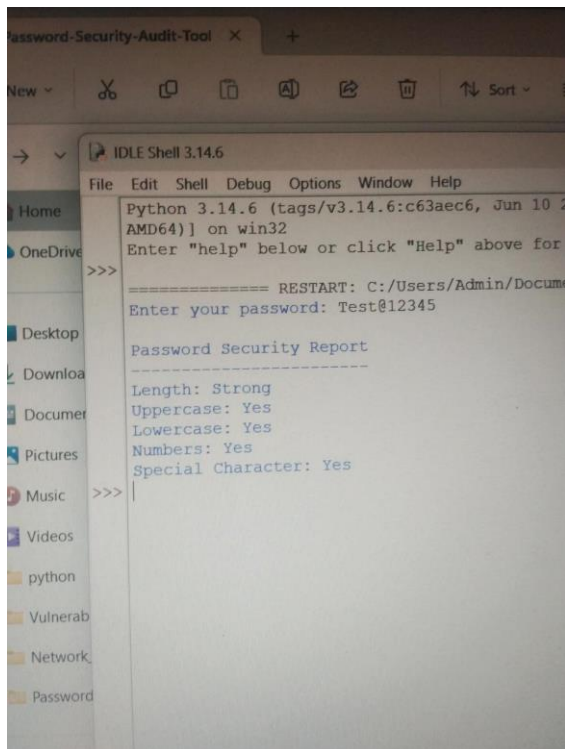
Library: bcrypt

File: common_passwords.txt

Platform: Windows

PASSWORD STRENGTH CHECKING:

The tool checks whether the password contains a sufficient number of characters and includes uppercase letters, lowercase letters, numbers, and special characters. These checks help identify whether a password follows basic password security requirements.



COMMON PASSWORD CHECKING:

The tool checks the entered password against a list of commonly used passwords stored in the common_passwords.txt file. If the password is found in the list, it is reported as a common and weak password.

```
File "C:/Users/Admin/Documents/password_audit.py", line 1
with open("common_passwords.txt", "r") as file:
FileNotFoundError: [Errno 2] No such file or directory:
>>>
===== RESTART: C:/Users/Admin/Desktop/password_
Enter your password: password

Password Security Report
-----
Length: Strong
Uppercase: No
Lowercase: Yes
Numbers: No
Special Character: No
Common Password: Yes - Weak
>>>
```

PASSWORD CRACK TIME ESTIMATION:

The tool estimates the approximate time required to crack a password using a brute-force approach. The estimation is based on the password length, available character types, and an assumed number of guesses per second. A stronger and longer password generally requires more time to crack.

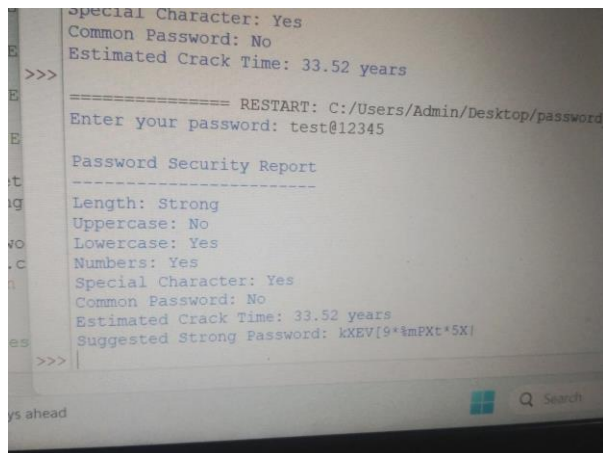
```
Lowercase: Yes
Numbers: No
Special Character: No
Common Password: Yes - Weak
>>>
===== RESTART: C:/Users/Admin/Desktop/pas
Enter your password: test@12345

Password Security Report
-----
Length: Strong
Uppercase: No
Lowercase: Yes
Numbers: Yes
Special Character: Yes
Common Password: No
Estimated Crack Time: 33.52 years
>>>
```

STRONG PASSWORD GENERATION:

The tool generates a random 16-character password using uppercase letters, lowercase letters, numbers, and special

characters. This helps users create stronger passwords that are difficult to guess or crack.

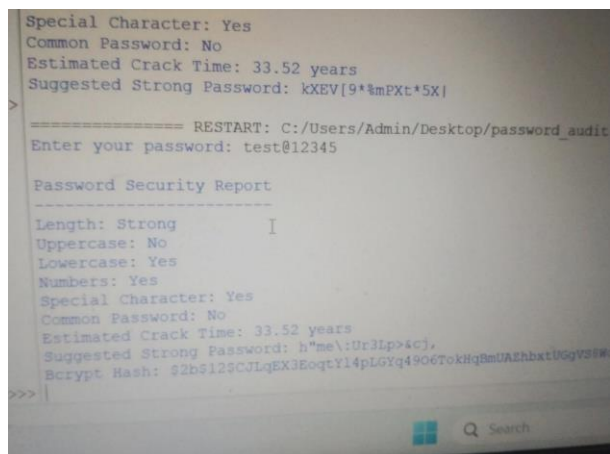


```
Special Character: Yes
Common Password: No
Estimated Crack Time: 33.52 years
>>>
===== RESTART: C:/Users/Admin/Desktop/password
Enter your password: test@12345

Password Security Report
-----
Length: Strong
Uppercase: No
Lowercase: Yes
Numbers: Yes
Special Character: Yes
Common Password: No
Estimated Crack Time: 33.52 years
Suggested Strong Password: kXEV[9*!mPXt*5X!
>>> |
```

SECURE PASSWORD HASHING USING BCrypt:

The tool uses the bcrypt hashing algorithm to securely protect passwords. Instead of storing the original password, bcrypt converts it into a salted hash. This improves password security and helps protect stored passwords from being exposed in plain text.



```
Special Character: Yes
Common Password: No
Estimated Crack Time: 33.52 years
Suggested Strong Password: kXEV[9*!mPXt*5X!
>
===== RESTART: C:/Users/Admin/Desktop/password_audit
Enter your password: test@12345

Password Security Report
-----
Length: Strong
Uppercase: No
Lowercase: Yes
Numbers: Yes
Special Character: Yes
Common Password: No
Estimated Crack Time: 33.52 years
Suggested Strong Password: h"me\;Ur3Lp>4c].
Bcrypt Hash: $2bs12$CJLqEX3EoqtYl4pLG7q4906T0khqBmUAZhbxt0qyvsW
>>> |
```

TESTING AND RESULTS:

The Password Security Audit Tool was tested with different passwords to verify its functionality. The tool successfully checked password strength, identified common

passwords, estimated crack time, generated strong password suggestions, and produced bcrypt hashes. The test results show that the tool performs the required password security checks successfully.

CONCLUSION:

The Password Security Audit Tool provides a simple and effective way to understand password security. It can identify weak and common passwords, estimate crack time, generate strong passwords, and securely hash passwords using bcrypt. This project helped in understanding the importance of strong passwords and secure password storage.